

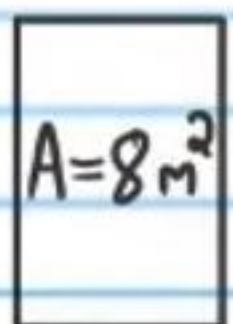
Pre-Algebra Practice Final Test-Version II

Instructions

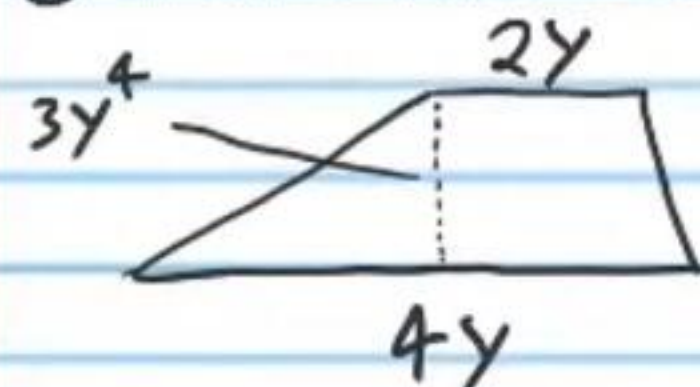
- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show all work.
- iii) Draw a rectangle around your answers.
- iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

① a) Find the average of the numbers below.

$$0.5x, 0.8x, 0.8x, 0.7x, 0.69x$$

b)  $A = 8 \text{ m}^2$ $L = ?$
 $W = 2 \text{ m}$

② Find the area of the trapezoid below.



Find the algebraic solutions and the graphical solutions of the following inequalities.

③ $-1 \geq 4 + \frac{z}{-3}$

④ $6.8 + 2.5x < 8 - 5.5x$

⑤ a) Find the Volume of a rectangular prism with length $\frac{2x}{7}$, width $\frac{3x^2}{5}$, and height $9x$.

b) Find the circumference of a circle with radius 5.99 feet. Use 3.14 for π .

⑥ a) If 22% of a number is 42, what is the number?

b) What percent of 8.5 is 3.4?

Solve.

$$\textcircled{7} \quad -2(x-3) + 4(2x+1) - 5x = 3(4-x) - (3x+2)$$

Graph the following inequality. Show all work and identify the points you use to graph the inequality. Label the graph properly.

$$\textcircled{8} \quad x \leq 2y + 3$$

$\textcircled{9}$ After an upgrade, the memory capacity of Kevin's computer was increased from 10 gigabytes to 27 gigabytes. What was the percent increase? You must write an algebraic equation and use it to solve this problem; otherwise, you will receive no points.

Solve.

$$\textcircled{10} \quad 8 + 3(2x+2) - 4 = 3x - 2(-4-2x) \quad \textcircled{11} \quad 3.5y + 1 = -8.6y - 4y + 1$$

⑫ Ronald put \$461 in a savings account that paid simple interest yearly. After leaving the money in this account for three years, it grew to \$543.98. Assuming he made no withdrawals or deposits, what was the interest rate of the account?

Solve the following equation and check your answer.

⑬ a) $-2(x-2) = 4x+19-3x-8x$ b) check

⑭ Use the equation below and the given values to find the unknown.

$$X = X_0 + V_0 t + \frac{at^2}{2}, \quad X=20, X_0=5, t=2, a=-1, V_0=?$$

Graph the following equation. Show all work and identify the points you use to graph the equation. Label your graph properly.

⑮ $-2x + 3y = 6$

⑯ a) Find A if $A = \pi r^2$, $r=4$, and $\pi=3.14$.

b) Find V if $V = V_0 + at$, $t=7$, $V_0=1$, and $a=3$.

Solve.

⑪ $7 - 2z - 3z + 1 + 10z = -4z + z - 3$ ⑫ $y + \frac{y}{4} - 2 = 3 - \frac{7y}{8}$

Evaluate.

⑬ $-y^2 + 4y^3 - 2y + 2(y+1)^2, y = -2$ ⑭ $9 + \frac{3}{z^2 + 1} - 3z, z = 0$

⑮ $6 - 3(x-2)^2 + 4x^2, x = -3$ ⑯ Evaluate your answer for problem #1a if $x = -3.5$.

⑰ $-3x^6y^3 \div \frac{24y}{25x}$

⑱ a) $\frac{-8a^5b^7}{-56a^8b^4}$

b) $\frac{0.48x^2y^8}{-0.03x^4y^7}$

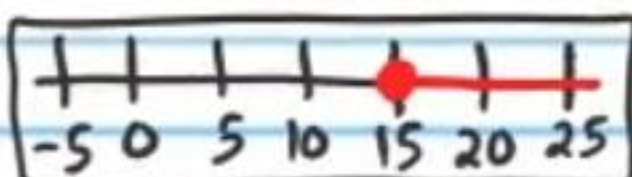
⑲ Find the perimeter of the triangle below.

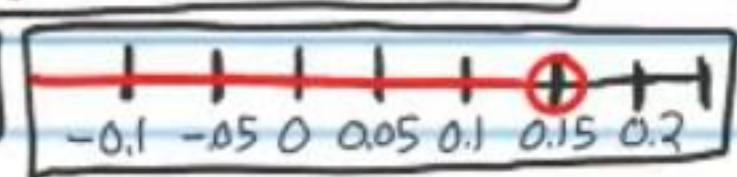


Pre-Algebra Practice Final Test-Version II Answers

① a) $0.698X$ b) $L = 4m$

② $9y^5$

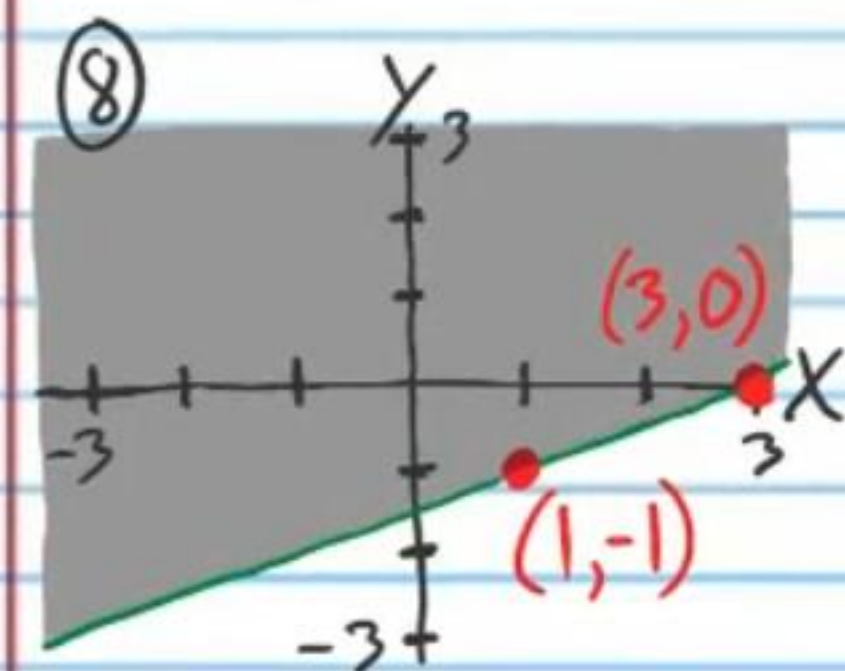
③ $Z \geq 15$  A number line from -5 to 25 with tick marks every 5 units. A solid red dot is at 15, and a red line segment extends to the right from this dot.

④ $X < 0.15$  A number line from -0.1 to 0.2 with tick marks every 0.05 units. An open red circle is at 0.15, and a red line segment extends to the left from this circle.

⑤ a) $54x^4 / 35$ b) 37.6172 ft

⑥ a) 190.90 b) 40%

⑦ $X = 0$



⑨ 170%

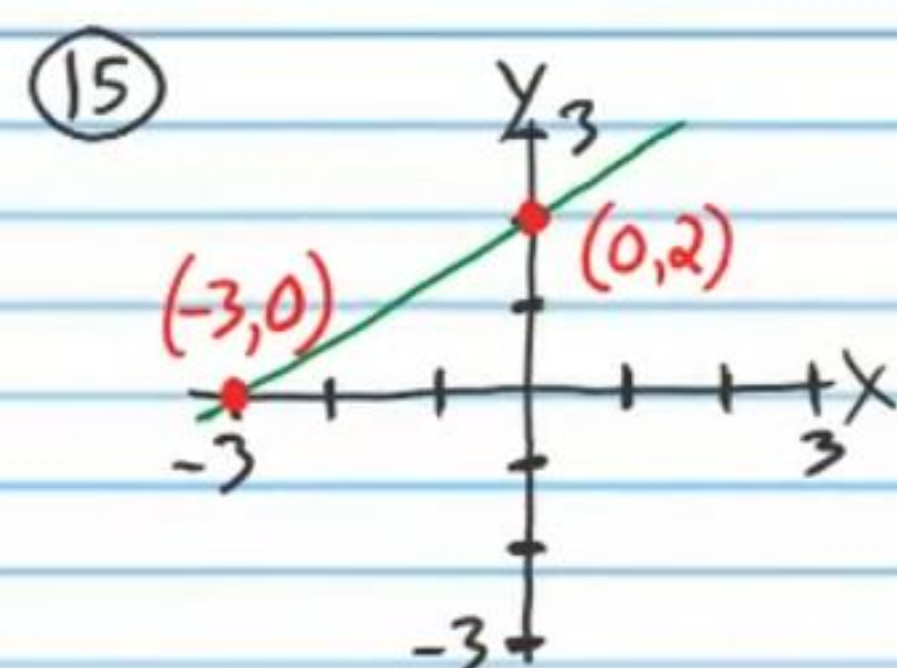
⑩ $X = 2$

⑪ $Y = 0$

⑫ 6%

⑬ a) $X = 3$ b) $-2 = -2$

⑭ $V_0 = 17\frac{1}{2}$



⑯ a) $A = 50.24$ b) $V = 22$

⑰ $Z = -11/8$

⑱ $Y = 40/17$

⑲ -30

⑳ 12

㉑ -33

㉒ -2.443

㉓ $-\frac{25x^7y^2}{8}$

㉔ a) $\frac{b^3}{7a^3}$ b) $-\frac{16y}{x^2}$

㉕ $24Z$